

Conditional Probability

Definition. For events A and B with $P(B) > 0$, the conditional probability of A given B is defined as

$$P(A \mid B) = P(A \text{ and } B) / P(B).$$

Requirement. The condition $P(B) > 0$ must be checked before the formula is applied. The denominator can never be zero.

Interpretation. Conditioning restricts the sample space to B and then rescales the measure so that the total mass is one again.

Chain rule. For three events with positive probability,

$$P(A \text{ and } B \text{ and } C) = P(A) * P(B \mid A) * P(C \mid A \text{ and } B).$$

Independence. A and B are independent exactly when

$$P(A \mid B) = P(A),$$

which requires $P(B) > 0$ as well.